

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions
- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

8. Model Generalization vs Overfitting

Performance:

Model Training Testing

A	98%	70%
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B	90%	88%
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Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

23. Multi-Model Comparison

Results:

odel	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Compare and evaluate the best overall model.

24. Motion Detection under Crowd Density

Results:

Method	Sparse Crowd	Dense Crowd
Optical Flow	High	Low
Motion Models	Moderate	High

Compare and evaluate best approach.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

28. Motion Model Evaluation

Results:

Model	Realism	Speed
A	High	Low
B	Moderate	High

Compare and evaluate based on application context.

Code of Conduct:

I K.ASWINI(192312397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application ; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

3. Optical Flow Reliability vs Noise Sensitivity

- **Method A:** Stable, less noise, but misses small motions
- **Method B:** Detects fine motion, but noisy

Comparison:

- Accuracy: B better for fine motion
- Robustness: A better (less noise)
- Reliability: A more consistent

Evaluation:

For **motion tracking**, especially in real-world scenarios (e.g., surveillance, robotics), **Method A is more suitable** because stable vectors ensure reliable tracking. Noisy vectors from Method B may lead to incorrect motion paths.

Conclusion:

Choose **Method A** for tracking stability

Choose **Method B** only when detecting very small movements is critical (e.g., medical imaging)

5. Motion Estimation vs Computational Cost

- **Method A:** 5% error, high computation
- **Method B:** 10% error, low computation

Comparison:

- Accuracy: A better
- Speed: B faster
- Resource usage: B efficient

Evaluation:

- **Real-time applications (e.g., video streaming, autonomous driving):**
 - **Method B** (faster, acceptable error)
- **Offline applications (e.g., video analysis, research):**
 - **Method A** (higher accuracy preferred)

Conclusion:

Real-time → **Method B**

Offline → **Method A**

8. Model Generalization vs Overfitting

Model Training Testing

A	98%	70%
B	90%	88%

Comparison:

- Model A: Overfitting (huge gap)
- Model B: Good generalization (small gap)

Evaluation:

- Model A memorizes training data → poor real-world performance
- Model B performs consistently on unseen data

Conclusion:

Model B is more reliable for deployment

Model A has high overfitting risk

21. Noise Impact on Detection

Noise Level Accuracy

Low	92%
Medium	80%
High	60%

Comparison:

- Accuracy drops significantly as noise increases

Evaluation:

- System is **sensitive to noise**
- Robust systems should maintain stable performance

Conclusion:

Current system has **low robustness**

Needs preprocessing (filtering, denoising) to improve performance

23. Multi-Model Comparison

	Model Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Comparison:

- A: Fast but weak
- B: Balanced
- C: Best accuracy but slow

Evaluation:

- **Best overall:** Model B (balance of all factors)
- C is ideal only when speed is not important

Conclusion:

Model B is the best overall choice

24. Motion Detection under Crowd Density

	Method	Sparse	Dense
Optical Flow	High	Low	
Motion Models	Moderate	High	

Comparison:

- Optical Flow struggles in dense scenes (occlusion)
- Motion Models handle crowd better

Evaluation:

- Sparse scenes → Optical Flow works well
- Dense scenes → Motion Models better

Conclusion:

Use **Motion Models** for crowded environments
Optical Flow only for simple scenes

27. Video Processing Pipeline Comparison

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Evaluation:

- Real-time → A (speed priority)
- Balanced → B (best trade-off)
- High precision → C (offline use)

Conclusion:

Pipeline B is most suitable overall

28. Motion Model Evaluation

Model	Realism	Speed
A	High	Low
B	Moderate	High

Evaluation:

- Model A: Accurate but slow → simulations, animation
- Model B: Faster → real-time systems

Conclusion:

Real-time applications → Model B

High realism needs → Model A
